

DONCASTER SHEFFIELD, United Kingdom

WMO: 034054

Lat: 53.475N

Lon: 1.004W

Elev: 55

StdP: 14.67

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.2	28.5	21.7	16.4	29.5	24.5	18.7	32.1	33.0	48.3	30.0	48.4	5.1	170	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	80.8	65.7	77.1	64.4	73.6	62.7	67.4	76.8	65.6	74.2	63.9	71.7	10.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	89.9	71.3	62.5	84.8	69.5	60.9	79.9	67.4	31.8	76.5	30.4	74.2	29.3	72.1	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.0	23.6	20.9	DB	20.7	87.4	6.7	3.6	15.9	89.9	12.0	92.1	8.3	94.1	3.5	96.7
			WB	20.2	70.9	6.4	2.4	15.6	72.6	11.9	74.0	8.3	75.4	3.6	77.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	40.6	41.1	44.2	48.7	54.0	59.4	63.7	62.4	58.1	52.5	45.4	41.8	(6)
	DBStd	9.73	5.49	6.02	5.47	5.24	5.30	4.62	4.63	4.06	4.45	5.31	6.03	6.63	(7)
	HDD50	1303	292	254	194	85	21	0	0	0	1	33	160	263	(8)
	HDD65	5174	755	670	646	489	342	178	82	105	212	388	587	720	(9)
	CDD50	1687	2	5	12	46	145	282	426	383	244	111	23	8	(10)
	CDD65	81	0	0	0	0	1	9	42	23	5	1	0	0	(11)
	CDH74	699	0	0	0	3	23	111	372	152	31	7	0	0	(12)
	CDH80	133	0	0	0	0	1	20	78	28	5	1	0	0	(13)
(14) Wind	WSAvg	10.0	11.0	10.7	11.1	9.9	9.8	9.2	8.8	9.6	9.3	9.6	10.3	10.9	(14)
(15) Precipitation	PrecAvg	22.6	1.9	1.4	1.7	1.8	1.9	2.0	1.9	2.1	2.1	1.8	2.0	2.0	(15)
	PrecMax	31.5	4.6	4.1	4.2	4.0	6.3	6.1	5.9	4.6	5.8	5.4	5.0	6.7	(16)
	PrecMin	15.2	0.4	0.0	0.3	0.2	0.4	0.2	0.4	0.2	0.4	0.3	0.8	0.3	(17)
	PrecStd	3.9	0.9	0.9	0.9	1.0	1.3	1.3	1.2	1.1	1.3	1.1	1.0	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	59.0	64.0	71.8	77.1	82.8	86.4	83.8	78.4	70.0	62.7	59.0	(19)
		MCWB	53.6	53.1	53.8	57.4	63.2	66.6	67.3	66.0	65.8	63.4	58.4	54.6	(20)
	2%	DB	53.8	55.3	59.2	66.0	72.2	77.1	82.1	78.1	73.0	66.0	59.3	56.9	(21)
		MCWB	50.1	50.4	52.1	55.1	60.8	63.7	66.3	65.2	63.4	61.0	55.9	53.3	(22)
	5%	DB	51.7	52.2	55.8	62.2	68.2	73.0	78.6	74.1	69.6	62.8	57.2	54.0	(23)
		MCWB	48.2	48.1	50.2	54.0	58.1	61.7	64.4	63.1	61.3	58.7	54.0	50.8	(24)
	10%	DB	49.7	50.2	53.5	59.0	64.5	69.7	74.7	71.5	66.4	60.8	55.0	51.8	(25)
		MCWB	46.9	46.8	48.1	52.0	55.7	60.5	63.2	61.5	59.2	56.7	51.8	48.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	53.1	55.3	60.0	65.0	69.3	70.1	68.3	67.1	63.7	58.8	55.2	(27)
		MCDB	57.3	56.9	60.2	68.6	74.7	79.7	80.4	77.5	74.8	69.6	62.1	57.9	(28)
	2%	WB	51.1	51.2	52.9	56.8	61.9	65.0	67.8	66.3	64.5	61.4	56.4	53.6	(29)
		MCDB	53.6	54.6	58.2	63.5	70.4	73.2	77.7	75.1	71.1	65.2	59.2	56.5	(30)
	5%	WB	48.9	49.2	51.0	54.4	59.4	62.8	66.0	64.6	62.4	59.3	54.5	51.8	(31)
		MCDB	51.1	52.3	55.7	60.9	66.7	70.5	75.2	72.5	67.6	62.4	57.1	54.6	(32)
	10%	WB	46.6	47.2	49.0	52.1	57.1	60.9	64.2	63.1	60.5	57.3	52.1	49.3	(33)
		MCDB	49.1	50.3	53.0	58.2	63.0	68.2	72.8	69.7	65.1	60.4	54.4	51.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	11.3	14.0	16.3	16.7	16.7	17.9	16.7	15.6	13.2	11.3	10.5	(35)
		MCDBR	11.9	12.7	18.0	21.3	23.2	23.8	24.5	21.2	19.3	14.3	12.6	11.8	(36)
		MCWBR	10.7	9.9	12.3	12.8	13.4	12.8	10.9	10.4	11.2	10.2	10.8	10.3	(37)
	5% WB	MCDBR	12.0	12.7	17.3	20.0	21.5	20.5	21.4	18.7	16.9	13.7	12.5	11.7	(38)
		MCWBR	11.0	10.3	12.6	13.0	13.0	12.5	11.1	10.4	11.1	10.3	10.9	10.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.324	0.340	0.371	0.399	0.399	0.404	0.405	0.391	0.371	0.352	0.325	0.311	(40)
	taud		2.347	2.385	2.296	2.232	2.271	2.283	2.288	2.347	2.410	2.442	2.402	2.323	(41)
	Ebn at Noon		203	240	255	263	269	268	266	263	254	233	202	183	(42)
	Edh at Noon		19	24	32	39	40	40	39	35	29	23	17	16	(43)
(44) All-Sky Solar Radiation	RadAvg		212	426	767	1186	1490	1541	1501	1236	912	520	260	169	(44)
	RadStd		20	49	92	144	99	146	157	81	83	45	29	18	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (71 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page